MILITARY SPECIFICATION
FOR
FUZE, TRAINING, ELECTRONIC TIME: M744

This specification is approved for use by the USA Armaments Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the FUZE, TRAINING, ELECTRONIC, TIME: M744 which is used for training personnel in the use of the FUZES, ELECTRONIC TIME: M587 and M724. This fuze is used with the Fuze Setter: M36.

1.2 Description. The M744 is a Digital Electronic Time Fuze. It consists of an electronic timer and contains hardware which is identical in form and electronic function to the fuzes M587/M724 and will be used for training personnel in the care and use of the FUZES, ET: M587 and M724. The overall length of the fuze is approximately 5.97 inches; the weight is approximately 1.61 ± 0.05 pounds. The training fuze does not contain a power supply or any explosive elements.

2. APPLICABLE DOCUMENTS

2.1 The following documents and documents cited therein of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-Q-50829 - General Quality Assurance Provisions for Proximity Fuzes and Related Components
MIL-S-46844 - Solder Bath Soldering of Printed Wiring Assemblies, Automatic Machine Type
MIL-S-50827 - Soft Soldered Electrical Connections for Conventional Weapons Including other Related Electronic Devices

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Armament Research and Development Command, ATTN: DRDAR-TST-S, Dover, NJ, 07801 by using the self-addressed Standardization Document Improvement Proposal (DD FORM 1426) appearing at the end of this document or by letter.

FSC 1390

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STANDARD

Military

MIL-STD-105 — Sampling Procedures and Tables For Inspection by Attributes

MIL-STD-1168 — Lot Numbering of Ammunition

MIL-STD-1235 — Single and Multiple Continuous Sampling Procedures and Tables For Inspection by Attributes

DRAWINGS

ARMY MATERIEL COMMAND

PL 11726806 — Parts List For FUZE, TRAINING, ELECTRONIC TIME: M744

EPL 11726806 — Special Inspection Equipment For FUZE, TRAINING, ELECTRONIC TIME: M744

8864492 -- Box, Ammo, Metal, For Artillery and Rocket Fuzes

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Samples. The contractor is responsible for all requirements for providing and testing of samples as prescribed herein. When this specification is assigned to a subcontractor, it shall be the prime contractor's responsibility to insure that sample requirements are met. The prime contractor shall notify the Government at the time of assignment to a subcontractor whether the testing will be performed by the prime or by the subcontractor, as well as the method of control by the prime contractor.

3.1.1 Comparison sample. Five fuzes (11726806) selected at random from each accepted lot, or as specified by the Government Procuring Activity, shall be shipped for evaluation within five working days after lot acceptance to the technical agency designated by the procuring activity.

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3.2 Construction. The device shall be constructed in accordance with the applicable drawings (PL 11726806) to the extent prescribed herein.

3.2.1 Materials. Materials shall be those specified by the applicable drawings.

3.2.2 Dimensions and technical notes.

3.2.2.1 Listed. Those dimensions and technical requirements listed in the Classification of Defects (4.3.5) are mandatory.

3.2.2.2 Unlisted. The contractor may proposed changes to characteristics shown on the drawings but not listed in the Classification of Defects, for the purpose of adapting the item to established manufacturing practices. Such proposals must be accompanied by evidence that the change does not affect the function of the item and that all requirements will be met. If the Government confirms the contention of the contractor, the change will be approved for the duration of the contract. In case of dispute, the characteristics of the drawings shall apply. Approval of a change under provisions of this paragraph does not relieve the contractor from establishing and maintaining an adequate quality assurance program as elsewhere required. Provisions of this paragraph shall not be used to obtain approval for use of discrepant material (i.e., produced before approval is obtained); or for design changes, which should be requested in accordance with change provisions of the contract document.

3.2.2.3 Interchangeability. While the dimensions and tolerances shown in the drawings and specifications will generally insure satisfactory products, the contractor is notified that some selection or matching may be required to obtain combinations which provide proper function or economic use of components. The contractor is responsible for selecting combinations of tolerances within the specified limits which satisfy his process needs and also meet the specified performance and fit requirements.

3.3 Performance requirement. The performance requirement contained herein is mandatory. Other performance requirements which may exist for the same components or assemblies in other end use specifications (e.g., MIL-F-48702 for M724 fuzes) are not required and units failing those other tests may be used in production under this specification.

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3.3.1 Electronic Operation. The fuze shall be settable (see 6.3.1) to any time from 0.2 to 199.0 seconds, in 0.1 second increments, or to point detonating using Fuze Setter: M36. The fuze shall also be capable of being interrogated (see 6.3.2) by Fuze Setter: M36.

3.4 Workmanship. All parts shall be manufactured and finished in a thoroughly workmanlike manner to insure satisfactory functioning and durability (See MIL-A-2550, Condition of Materials, Parts and Assemblies).

3.4.1 Soldering. Soldering shall be in accordance with MIL-S-50827 or MIL-S-46844.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor quality assurance system. The contractor shall provide and maintain a quality assurance system in compliance with MIL-Q-40829.

4.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification in compliance with MIL-Q-50829.

4.3 Acceptance inspection. Inspection shall be as specified in MIL-A-2550 and in this document. The contractor shall notify the Government QA representative prior to proceeding with inspection for Classification of Defects (4.3.5). Inspection may be made at a higher assembly level provided that equal assurance provisions are satisfied, but not at a lower level than specified.

4.3.1 Lot formation. A lot shall consist of devices produced at one manufacturing location under the following conditions: continuous production; stabilized methods and techniques; and the same revision of drawings and specification except for those introduced without obsolescence.

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4.3.2 Sampling. Sampling shall be in accordance with MIL-STD-105 and this document. Unless otherwise specified, inspection level II shall apply. Where appropriate and when approved by the Government the contractor may use MIL-STD-1235. When any other than the specified lot size, for inspection purposes, is agreed upon in conformance with the Military Standards MIL-STD-105 or MIL-STD-1235, appropriate AQL's for the new inspection lots shall be selected to give the same consumer's risk as the established AQL's for the original lot size.

4.3.3 Lot acceptance. Lots shall be accepted on the basis of Classification of Defects (4.3.5) and Material and Component Certification (4.3.6).

4.3.4 Data recording. A written record shall be made of the results of all examinations and tests performed as specified elsewhere in section 4 of this specification. The test data shall be recorded on 80 column standard data processing punch cards, or other computer compatible input medium approved or specified by the Government. The defect summary shall be recorded in the inspection of the classification of defects. Data layout of the card shall be as approved or specified by the Government. The completed cards shall be forwarded to the designated Government agency within two working days after the Government acceptance of the item or lot. If inspections are conducted on a sublot basis, the acceptance data for each sublot shall be so identified. (Engineering information, section 6).

4.3.5 Classification of defects. The assembly shall be inspected by using the Classification of Defects and applicable drawings.

4.3.5.1 Fuze, ET: M744 (11726806).

Categories	Defects	AQL	Method of inspection
Critical	None defined		
Major			
101	Error on interrogation or .745 dia to center contact less than min	100 percent screening	4.6.2 (Gage)

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Categories	Defects	AQL	Method of inspection
102	2.00-12-UNS-1A thread damaged	.65	Gage
103	Nose cone improperly crimped to sleeve	.65	Visual DS-1
104	Seating shoulder to forward exceeds maximum	.65	Gage
105	Booster cup diameter or true position exceeds maximum	.65	Gage
106	Wrench slot max span or min width improper	.65	Gage
107	Seating shoulder to aft end excess maximum	.65	Gage
108	"INERT" marking, blue band or holes in sleeve missing	.65	Visual
109	Setting ring missing or damaged	.65	Visual
Minor			
201	Lot No. missing, illegible, or improper	4.0	Visual

4.3.5.3 Shipping and Storage Containers.

Categories	Defects	AQL	Method of inspection
Minor			
201	Dessicant bag missing	1.5	Visual
202	Package marking missing or illegible	4.0	Visual
203	Fuze missing	1.5	Visual

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4.3.5.4 Defect standards.

DS-1 - The nose cone shall be tight in the sleeve with the sleeve crimped over evenly against the nose cone for a full 360°. There shall be no evidence of removal of the cadmium plating or bare steel visible on the sleeve when viewed with the naked eye. Removal of the chromate coating exposing the cadmium plating is permitted.

4.3.6 Material and component certification. Prior to acceptance under this procurement, the contractor shall demonstrate by means of certifications or statement of findings, that only materials and components conforming to the item specification and drawings, have been used.

4.4 Test conditions and equipment.

4.4.1 Test conditions. Unless otherwise specified, all measurements are to be made at normal ambient room conditions of temperature, relative humidity, and altitude. Temperature conversion between Fahrenheit and Celsius shall be carried to the nearest whole degree. The following table of tolerances shall be applied where applicable to specified test conditions unless otherwise defined in the test procedures:

a.	Temperature in test chamber	±5°F
b.	Relative humidity	±5 percent
c.	Vibration amplitude	±10 percent
e.	Electrical and physical measurements	±10 percent

4.4.2 Inspection equipment. Inspection equipment shall be in accordance with the design level shown by EA 11726806 and of sufficient accuracy and quality to permit performance of the required inspection. Gage and instrument setting shall be a function of the equipment, and independent of the operator. Equipment shall be marked to indicate the limit of acceptance for each test, but shall not be so marked as to reveal classified security information. Test procedures and calibration procedures shall conform to MIL-Q-50829.

4.5 Test procedures.

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4.5.1 Setting test. The fuzes shall be set to the settings specified below using the M36 Fuze Setter. The fuzes shall then be interrogated by the setter and the setter display noted. If on any operation the setter displays an "E" (for error) the operation shall be repeated. If the requirements of this paragraph are met on the next three consecutive operations, the initial result shall be ignored. This provision shall not be used to pass a fuze which operates intermittantly.

Setting	Interrogation Result
PD	P
0.2 seconds	0.19 to 0.22 inclusive
100.0 seconds	99.94 to 100.08 inclusive
199.9 seconds	199.84 to 199.98 inclusive

5. PREPARATION FOR DELIVERY

5.1 Packing and packaging. The fuzes shall be packed and marked for shipment as specified on drawing 8864492.

5.2 Records.

5.2.1 Ammunition data cards. Data card information shall be as specified in MIL-STD-1167. A sample data card will be provided by the Government procuring activity.

6. NOTES

6.1 Items to be included in the procurement documents.

6.1.1 Documents. Procurement documents should specify the title, date and number of this specification.

6.1.2 Samples. The number of units should be increased to include the samples as required in 3.1 since they are not included in the number of units required for delivery.

6.1.3 Data requirements. The following data are required by this specification to be furnished to the Government and should be entered on DD Form 1423 for each contract:

Examination and Test Records on IBM Cards (4.4.4)
Ammunition Data Cards (5.2.1)
MIL-Q-50829 -- Contractor-furnished Designs (3.6.1)

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6.2 Engineering information. Upon request, copies of the data collected in the acceptance of a lot will be made available to the Government representative. Additional data or data requiring special requirements, when needed for engineering purposes, will be specifically defined by the contract.

6.3 Definitions.

6.3.1 Set. A fuze is set to a time or PD as indicated on Fuze Setter: M36.

6.3.2 Interrogate. The fuze setter displays the time or PD mode that has been previously set into the fuze.

6.4 MIL-Q-50829. Copies of MIL-Q-50829 may be obtained from the Harry Diamond Laboratories, ATTN: Contracting Officer, 2800 Powder Mill Rd; Adelphi, MD. 20783

6.5 Specification and modification. Prior to production, modifications or changes to tolerances or definitions in specifications, e.g., MIL-S-46844, MIL-STD-331, etc should be delineated.

6.6 Note to procuring agency. This specification is written assuming that M744 training fuzes will be procured simultaneously with M587 or M724 electronic time fuzes. If M744 fuzes are to be separately procured, consideration should be given to additional requirements, such as a First Article Approval Sample.

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Army -AR

Preparing Activity:
Army - AR

Agent:
Army - HD

Project No: 1390-A282

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